

EN-CHI LEE

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SCHOOL College of Science, University of Arizona, Tucson, Arizona

 GitHub  ORCID  Website  LinkedIn

EDUCATION

2027– **California Institute of Technology**, Pasadena, California
PhD in Geophysics

2023–2026 **University of Arizona**, Tucson, Arizona
BSc in Geosciences (emphasis in Geophysics) & **BSc in Planetary Geoscience**
Minors: Mathematics & Oceanic and Atmospheric Sciences

2022–2023 **National Taiwan University**, Taipei, Taiwan
Major: Geosciences
Transferred to the University of Arizona

RESEARCH EXPERIENCES & WORKSHOPS

2026–PRESENT **Senior independent study**
Department of Geosciences, University of Arizona
Mentor: Dr Amanda Hughes

- Developing workflows to generate geologically consistent synthetic reflection seismic data.
- Implementing latent space diffusion model to perform seismic inversion and formal uncertainty quantification.

2025–PRESENT **Yin Climate Research Group**, Undergraduate researcher
Department of Geosciences, University of Arizona
Mentor: Dr Jianjun Yin

- Analysing tide gauge data and decomposing the relative contributions of different sea-level change mechanisms to coastal flooding.
- Modelling the effect of atmospheric forcing on local sea-level changes.

2024–PRESENT **Arizona Polar Ice Lab**, Undergraduate researcher
Department of Geosciences, University of Arizona
Mentor: Dr Christopher Harig

- Developed codebase *ULMO* to analyse satellite and oceanographic data; used it to jointly estimate and model sea-level budgets in ocean basins.
- Showed that mis-modelled glacial isostatic adjustment (GIA) is a significant source of discrepancy in previous sea-level budget estimates.

2025 **Geology Field School**
Department of Geosciences, University of Arizona
Instructors: Dr Peter DeCelles, Dr Barbara Carrapa, & Dr Paul Kapp

- Mapped several faults and related structures in the Sevier Thrust Belt and Laramide Orogeny.
- Constructed map-scale and regional cross-sections to understand the tectonic evolution of the North American Cordillera.

2023 **Dynamics and Modeling Laboratory**, Undergraduate research intern
National Center for Theoretical Sciences / Department of Atmospheric Sciences, National Taiwan University
Mentor: Dr Hung-Chi Kuo

- Implemented numerical solver for dynamical boundary layer adjustment and similar PDE systems.
- Evaluated the performance of the Pangu AI model in forecasting typhoon development compared to numerical weather prediction; showed the model was unstable and yielded nonphysical storm structures for longer forecasts.

2022

Seismology Skill Building Workshop

IRIS Consortium (now EarthScope)

- Participated in the self-motivated workshop with a 99% average score.
- Performed analysis using Shell scripts, SAC, and ObsPy to study the behaviours of seismic events, earthquake positioning, and waveforms.

SELECTED PUBLICATIONS

- [4] Yin, J & **E-C Lee** (2026). High-tide and storm-surge flooding along the U.S. East and Gulf Coasts. In preparation for *Journal of Climate*.
- [3] **Lee, E-C** & C Harig (2025). The impact of GIA corrections on gravimetric basin-scale ocean mass budgets. Under review at *JGR: Solid Earth*; preprint available on *EarthArXiv*. DOI: [10.31223/X5RR16](https://doi.org/10.31223/X5RR16)
- [2] **Lee, E-C** (2025). Unified localisation of mass over oceans (ULMO) [Software]. *Zenodo*. DOI: [10.5281/ZENODO.17625698](https://doi.org/10.5281/ZENODO.17625698)
- [1] **Lee, E-C** & C Harig (2024). Investigating the variance of GIA corrections on gravimetric basin-scale ocean mass budgets [Abstract]. *AGU24 Fall Meeting*. ADS: [2024AGUFMG51B.3394L](https://ui.adsabs.org/2024AGUFMG51B.3394L)

SELECTED PRESENTATIONS

- MAY 2025 **Investigating the variance of GIA corrections on gravimetric basin-scale ocean mass budgets (talk)**
Arizona Collaborative Consortium for Earth and Space Science Workshop, Northern Arizona University
- MAY 2025 **Geospatially conditioned diffusion for worldbuilding and fantasy cartography (talk)**
Mathematics of Generative AI Course Project Showcase, University of Arizona
- MAR 2025 **Investigating the variance of GIA corrections on gravimetric basin-scale ocean mass budgets (talk)**
GeoDaze 2025, University of Arizona
- DEC 2024 **Investigating the variance of GIA corrections on gravimetric basin-scale ocean mass budgets (poster)**
AGU24 Fall Meeting, Washington, DC
- SEP 2024 **Estimating basin-scale ocean mass budgets with satellite gravity (talk)**
McNair Scholar Research Conference, University of New Mexico
- AUG 2024 **Estimating basin-scale ocean mass budgets with satellite gravity (talk & poster)**
Undergraduate Research Opportunities Consortium Colloquium, University of Arizona
- MAY 2024 **Dust, habitability, and the evolution of the martian climate (talk)**
Geology of the Solar System Course Project Showcase, University of Arizona

EMPLOYMENT & SERVICES

2026

GEOS304 Structural Geology, Preceptor

Department of Geosciences, University of Arizona

- Review weekly lecture material for the lab component of the course.
- Create visual aids in Python to help students understand relevant concepts.
- Assist in lab instruction and provide real-time feedback to students.

- 2024–2026 **Arizona Journal of Interdisciplinary Studies**, Editor
University of Arizona
- Peer-review and edit interdisciplinary research articles for publication.
 - Co-organise outreach events and showcases of published research.
- 2024 **AGU24 Annual Meeting**, Student volunteer
Washington, DC

SELECTED HONOURS & DISTINCTIONS

- 2026 **Excellence in Undergraduate Research Award**
Department of Geosciences, University of Arizona
- 2026 **Thomas R Brown Foundation Distinguished Graduate Fellowship in Integrative Science**
College of Science, University of Arizona
Awarded 7 500 USD/year for intellectual merit, demonstrated academic excellence, and exceptional potential to advance knowledge in climate change.
- 2023–PRESENT **Dean's List with Distinction**
College of Science, University of Arizona
Awarded for a 4.0 GPA.
- 2023–PRESENT **Scholarship for Students Participating in International Science Olympiads and International Science Exhibitions to Study Abroad**
Ministry of Education, Taiwan
Awarded 30 000 USD/year to study abroad.
- 2023–2025 **Transfer Global Wildcat Award**
University of Arizona
Awarded 11 000 USD/year for outstanding academic performance.
- 2022–2023 **Fu Bell Scholarship**
National Taiwan University
Awarded 200 000 NTD/year for outstanding academic performance.
- 2022–2023 **Presidential Award**
National Taiwan University
Awarded for maintaining a top 5% GPA within the Department of Geosciences.
- 2021 **International Earth Science Olympiad**, Gold medallist
International Geoscience Education Organisation
Ranked second place globally among 185 contestants from 27 countries.

PROFESSIONAL AFFILIATIONS & MEMBERSHIPS

- 2026–PRESENT **Arizona Geological Society**, Member
- 2024–PRESENT **American Geophysical Union**, Member
- 2022–PRESENT **Seismological Society of America**, Member
- 2023–2026 **Arizona Science, Engineering, and Mathematics Scholars (ASEMS)**, Scholar